Fiscal Year:	FY 2026	**Task Last Updated:**	FY 03/25/2026
PI Name:	Davis, Anthony Ph.D.		
Project Title:	Evaluation of the Cellular and Molecular Responses of Lung and Liver Cells Derived From iPSCs With Varying Intrinsic Radiosensitivity to Space Radiation		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2020-2021 HERO 80JSC020N0001-HHCSR, Omnibus2. Human Health Countermeasures and Space Radiation Topics Appendix C; OMNIBUS2-Appendix D
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No. of Post Docs:	0	**No. of PhD Degrees:**	2
No. of PhD Candidates:	0	**No. of Master' Degrees:**	1
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
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Key Personnel Changes/Previous PI:			
COI Name (Institution):	Wu, Jun Ph.D. (University of Texas Southwestern Medical Center at Dallas) Story, Michael Ph.D. (University of Texas Southwestern Medical Center at Dallas)		
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Task Description:	Deleterious mutations in genes required for the DNA damage response (DDR) confer radiosensitivity and are associated with increased cancer risk. However, variability in intrinsic radioresponse across the general population is likely driven by factors beyond overt loss of DDR function, including genetic variation, epigenetic regulation, metabolic state, and subtle perturbations in DDR pathways. A key unanswered question is whether intrinsic radiosensitivity is associated with cancer risk following radiation exposures on Earth and during deep space missions. To address this, we will leverage a novel, genetically diverse panel of primary human skin fibroblasts (PHSFs) derived from 90 individuals that exhibit an approximately fourfold range in radiosensitivity to γ-radiation. These PHSFs will be used to generate hTERT-immortalized fibroblasts, induced pluripotent stem cells (iPSCs), and iPSC-derived hepatocytes and lung epithelial cells, while preserving the intrinsic radiosensitivity of the parental lines. Lung and liver cell types were selected due to their relevance as organs at risk for space radiation–induced lethal cancers. All experiments will incorporate both low–linear energy transfer (LET) radiation (γ-rays) and high-LET radiation, including a 5-ion galactic cosmic ray (GCR) simulation and representative component ions. We will apply a comprehensive suite of assays, including clonogenic survival, DNA repair capacity, DDR signaling, genomic stability, mutation frequency, cellular transformation, and oncogenic potential, complemented by transcriptomic, whole-genome sequencing, DNA methylation, and RNA-seq analyses. The overall goal is to identify molecular and genetic determinants of intrinsic radioresponse that are associated with susceptibility or resistance to space radiation–induced oncogenic transformation and carcinogenesis. The ability to identify individuals at elevated or reduced risk following low- and high-LET radiation exposure would provide a critical foundation for risk stratification and mitigation in future space missions.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Deleterious mutations in genes required for the DNA damage response (DDR) cause radiosensitivity and are associated with increased cancer risk. However, variability in intrinsic radioresponse across the human population is likely governed by other factors besides loss of function of DDR proteins, including genetics, epigenetics, metabolic pathways, and slight perturbations in the DDR. A key unanswered question is if a relationship between intrinsic radiosensitivity and risk for cancer from radiation exposures on earth and in deep space exist. To address this, we will use a panel of 90 primary human skin fibroblasts (PHSFs) that have a four-fold difference in radioresponse to gamma-rays to generate immortalized PHSFs and iPSCs and iPSC-derived hepatocytes and lung epithelial cells generated from the PHSFs. We will use this panel of PHSFs and derivatives from these parental cells (hTERT immortalized, iPSC, and iPSC-derived lung epithelial cells and hepatocytes) to mechanistically examine the role that intrinsic radiosensitivity plays in the cellular response to low (gamma-rays at UT Southwestern) mixed (5 ion Galactic Cosmic Ray simulation (GCRsim) and representative (component ions of the GCRsim at the NSRL) radiation. These experiments will allow us to characterize how inter-individual radioresponses contribute to the response to mixed and individual species of the space radiation environment and its impact on radiation-associated cancer risk, as monitored by genomic instability, which is a hallmark of cancer. The ultimate goal is to identify candidate biomarkers to determine individual radiation susceptibility/resistance to radiation, in particular space radiation. We believe this will result in initial recommendations for clinical practice guidelines incorporating biomarkers into standard measures for future space missions.
Task Progress:	The response to ionizing radiation (IR) varies among individuals. This variability is well established in individuals with deleterious mutations in DNA damage response (DDR) genes, which confer radiosensitivity and increased cancer risk. However, differences in intrinsic radioresponse are also observed in individuals without overt DDR defects, suggesting that additional factors—including genetic variation, epigenetic regulation, metabolic state, and subtle perturbations in DDR pathways—contribute to radiation response. A key unanswered question is whether intrinsic radiosensitivity is associated with cancer risk following radiation exposures on Earth and during deep space missions. Conversely, it remains unclear whether individuals with greater intrinsic radioresistance are less susceptible to radiation-induced carcinogenesis. To address these questions, we will leverage a genetically diverse panel of more than 90 primary human skin fibroblast (PHSF) lines exhibiting an approximately fourfold range in radiosensitivity to γ-radiation. This panel includes donors of different sexes, ethnic backgrounds, and ages (childhood through late adulthood), providing a powerful platform to examine inter-individual variability. From these cells, we are generating hTERT-immortalized fibroblasts, induced pluripotent stem cells (iPSCs), and iPSC-derived hepatocytes and lung epithelial cells. Lung and liver models were selected due to their relevance as organs at risk for high atomic number and energy (HZE) particle-induced cancers. This system will enable mechanistic evaluation of how intrinsic radiosensitivity influences radiation-induced carcinogenesis following both low- and high-linear energy transfer (LET) exposures, including heavy ions relevant to deep space. We will assess endpoints including clonogenic survival, DDR signaling, DNA repair, genomic instability, mutation frequency, and cellular transformation, complemented by genomic and epigenomic analyses. We anticipate that these studies will identify molecular determinants of intrinsic radioresponse that may serve as biomarkers of oncogenic risk. The primary objectives are: Aim 1. Determine the relationship between intrinsic radioresponse and DDR signaling, radiation-induced genomic instability, mutation frequency, and cellular transformation across PHSFs, iPSCs, and iPSC-derived hepatocytes and lung epithelial cells following low- and high-LET radiation. Aim 2. Identify molecular and genetic features of intrinsic radioresponse associated with susceptibility to HZE-induced cellular transformation and carcinogenesis. To date, we have generated and validated 15 iPSC lines and continue to expand this resource. Importantly, iPSCs generally retain the radioresponse of their parental PHSFs following reprogramming. DDR signaling is also largely conserved, with radiosensitive lines exhibiting more sustained DDR activation and delayed double-strand break repair. In addition, IR-induced genomic instability appears elevated in radioresensitive PHSFs. We have initiated differentiation of iPSCs into liver-like cells and observe that their radioresponse is consistent with that of the parental lines. Overall, progress to date indicates that the project is on track. Continued support will enable completion of the proposed aims and advancement of biomarker development to assess individual susceptibility to radiation, particularly in the context of space radiation exposure.
Bibliography Type:	Description: (Last Updated: 04/30/2025)